

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 6

INDIAN SPACE RESEARCH ORGANISATION - 4

GAGANYAAN MISSION

- The Indian Human Spaceflight Programme was created by ISRO to develop the technology needed to launch a crewed spacecraft into low Earth orbit of 300-400 km.
- The first crewed flight is planned for December 2021 on indigenously built GSLV Mk-III rocket.
- ISRO has already developed most of the technologies required for crewed flight like re-entry mission capability, crew escape system, crew module configuration, thermal protection system and the prototype of a space suit.



GAGANYAAN MISSION

- In 2018 ISRO conducted a Crew Module Atmospheric Re-entry Experiment and a Pad Abort Test for the mission.
- The Gaganyaan project will cost less than Rs. 10,000 crore.
- The spacecraft is being designed to carry three people, and the 3.7-tonne capsule will orbit the Earth at 400 km altitude for a maximum time up to seven days.
- This mission will be remarkable as India and would be the 4th country in the World to fly in space with indigenously developed spacecraft after Russia, the U.S and China.

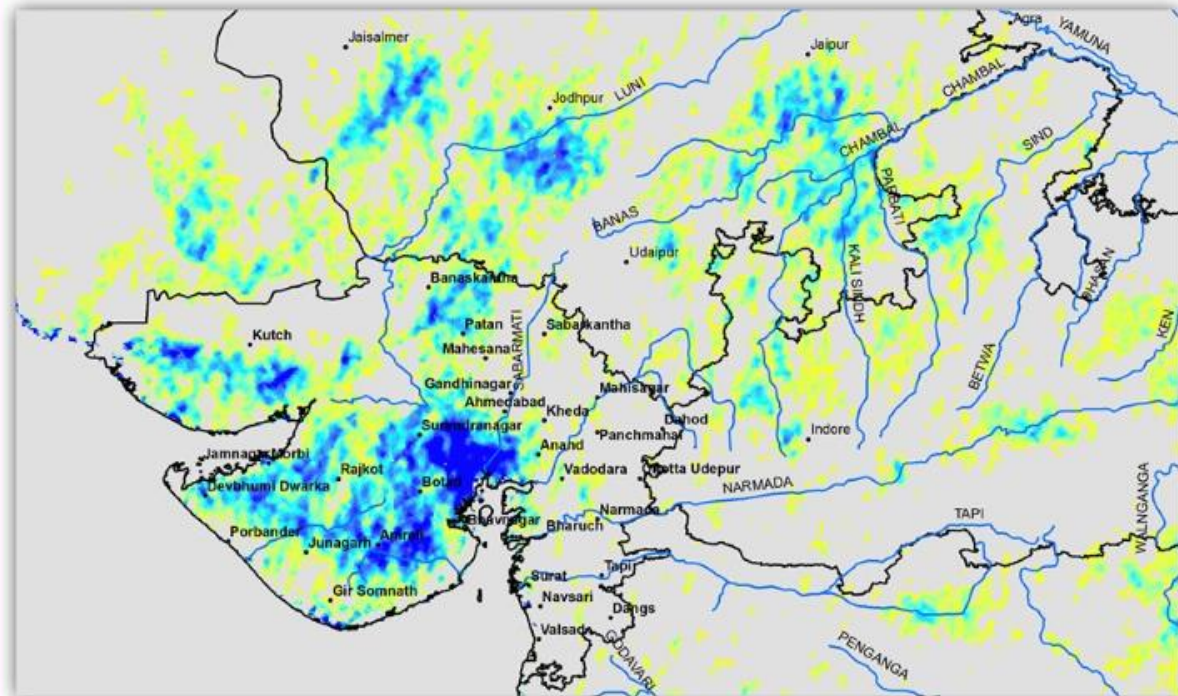


REMOTE SENSING

It is the technique of obtaining information about an object or phenomenon without making physical contact with the object. It is opposite to on-site observation.

Areas where remote sensing is used:

- Geography
- Land Survey and Mapping
- Oceanography
- Forest survey and Mapping
- Urban Planning
- Disaster Management
 - ✓ hazard assessment
 - ✓ tracking of hurricanes and
 - ✓ monitoring floods etc.
- Military and Intelligence gathering

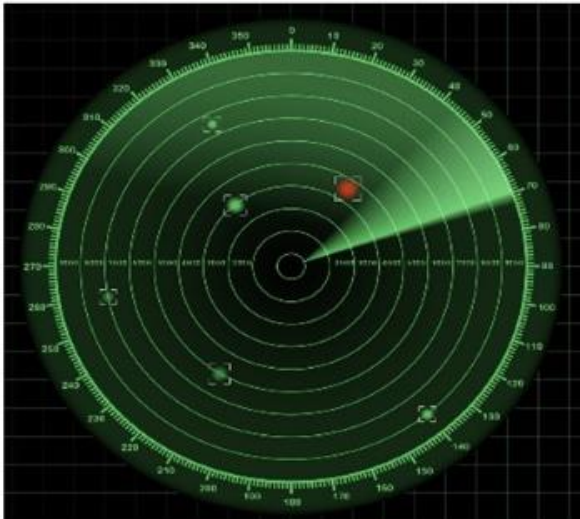


REMOTE SENSING

Some technologies that work on the concept of Remote Sensing are:

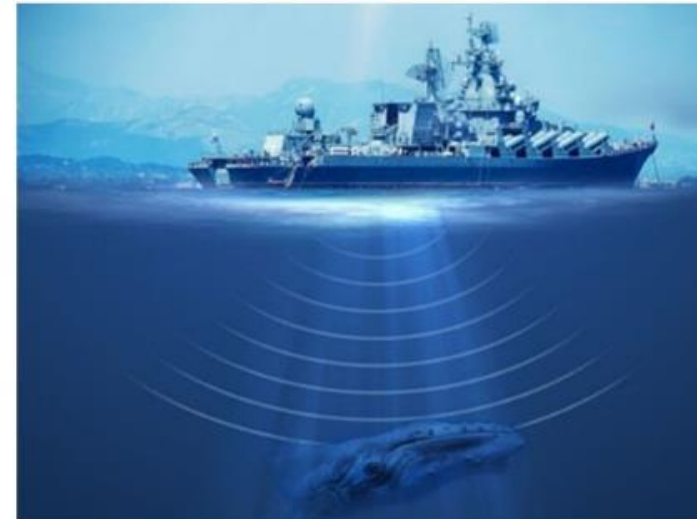
1 RADAR (Radio Detection and Ranging)

- ✓ It is an object-detection system that uses radio waves to determine the range, angle, or velocity of objects.



2 SONAR

- ✓ It is a technique that uses sound waves, usually underwater, to navigate, communicate with or detect objects on or under the surface of the water.

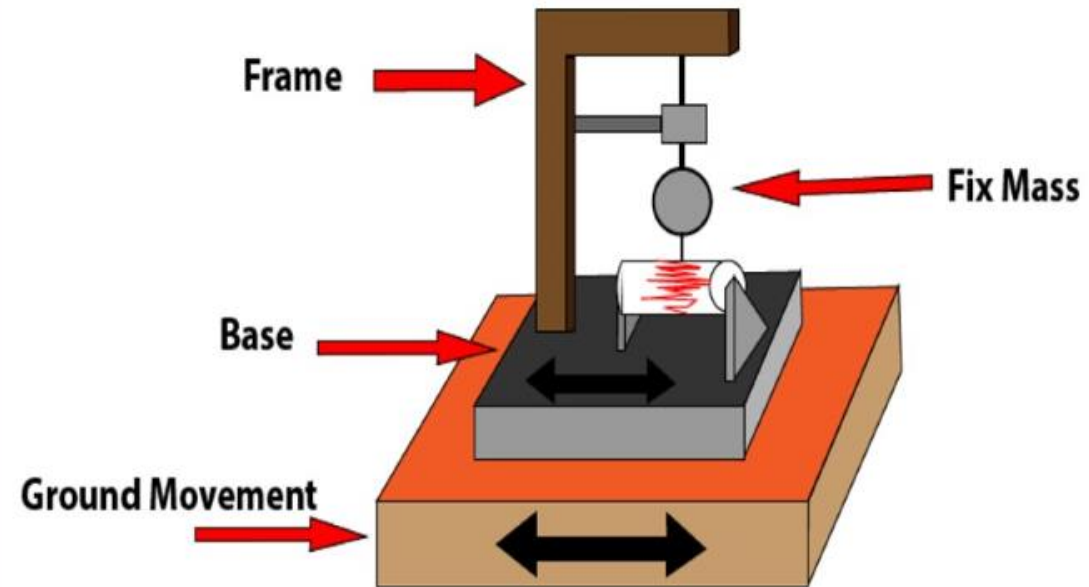


REMOTE SENSING

Some technologies that work on the concept of Remote Sensing are:

3 LIDAR

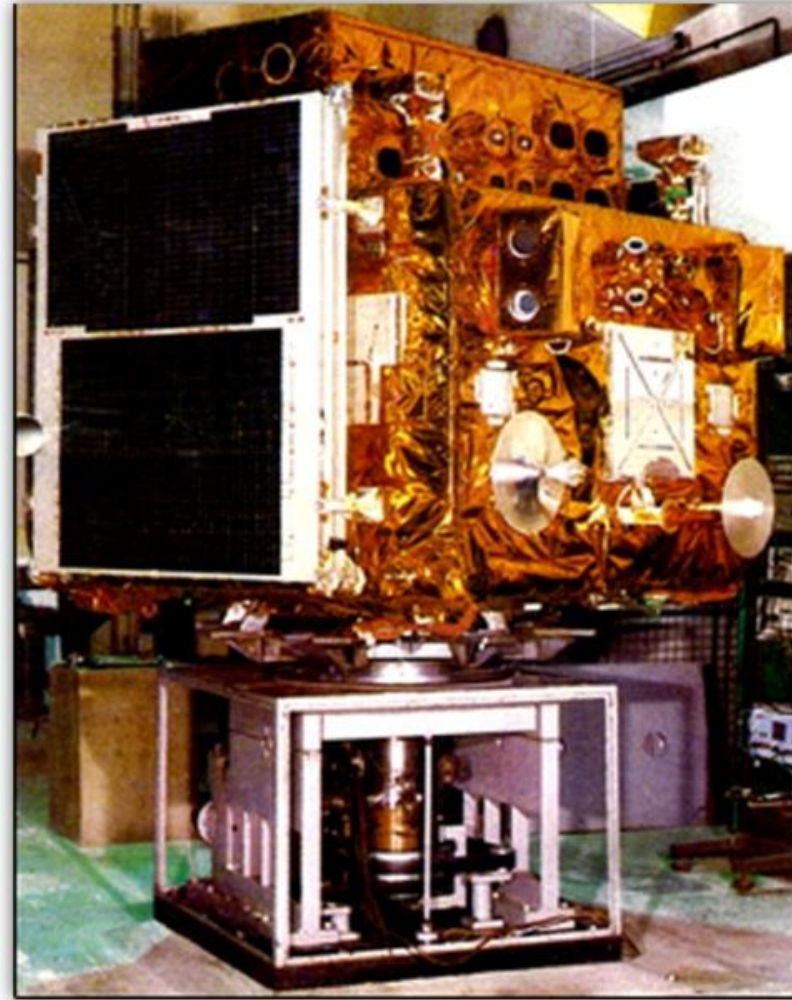
- It stands for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure distances.

4 Seismograms**5** Photometers**6** Earth Observation Satellites

Seismograph

INDIAN REMOTE SENSING PROGRAMME

- India's remote sensing programme under the Indian Space Research Organization (ISRO) started off in 1988 with the IRS-1A, the first of the series of indigenous remote sensing satellites, which was launched into a polar orbit in 1988 from a Soviet Cosmodrome.
- IRS-1A was followed by the launch of IRS-1B, an identical satellite, in 1991.
- These two satellites of the IRS series have been the workhorses for generating a lot of information in a variety of areas, such as agriculture, forestry, geology and hydrology etc.

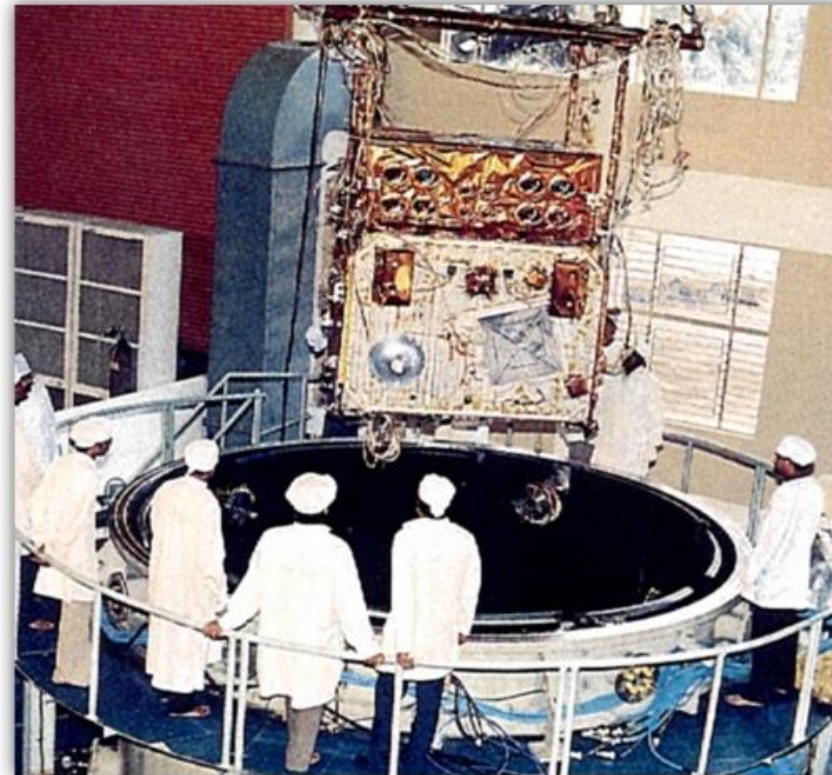


IRS-1A



INDIAN REMOTE SENSING PROGRAMME

- Currently 11 IRS satellites are in operation, which is the largest civilian remote sensing satellite constellation in the world.
- The management of these satellites and utilization of data from these satellites is done by National Natural Resources Management System (NNRMS), which is the nodal agency for natural resources management in the country.



INDIAN REMOTE SENSING PROGRAMME

Some satellites with specific purposes in IRS series are:

- Land/water resources applications
 - ✓ RESOURCESAT series and
 - ✓ RISAT series.
- Ocean/atmospheric studies
 - ✓ OCEANSAT series
 - ✓ INSAT-VHRR
 - ✓ INSAT-3D and
 - ✓ SARAL
- Large scale mapping applications
 - ✓ CARTOSAT series.



RESOURCESAT 2A

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